

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Robots will start to look more human!



Learning Outcome

Explain about technology and technical problems in communicative situations.

Ask your partner:

- What type of robotic pet would you consider having? Why?
- Have you seen any robotic pets online or on TV? What did they look like?

Can Do Statements

- I can talk about technology and gadgets.
- I can talk about recent actions and routines.
- I can express complex statements using words like *who*, *that* or *which*.

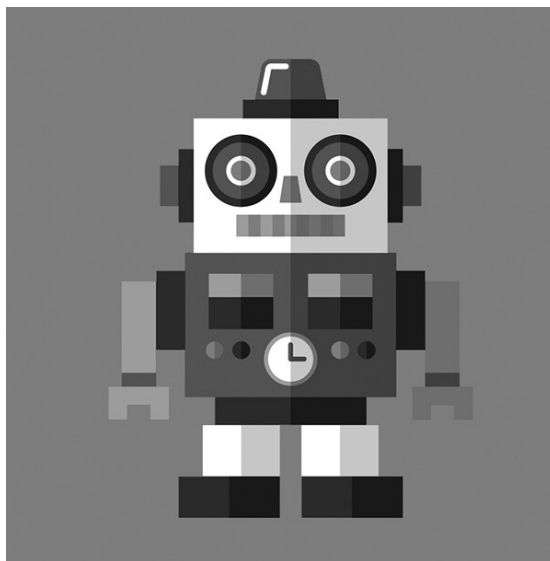




Let's Explore!

Most robots **seem cold*** to us.

* When we say a robot or person "*seems cold*" we mean that they do not display emotions or feelings.



Instructions: Read the text aloud.

Research into how robots function and their potential uses has grown substantially since the early 20th century. In 1939, Westinghouse built 'Elektro', who was humanoid in appearance, for the World's Fair. The robot, which relied on motors, photoelectric cells, relays and record players, performed 26 routines.

Since then, robotic engineers have been crafting robots to look, feel and act more human. They have been hoping this will make it easier for us to warm to machines that we see as cold.

They have been working toward creating robots that sense your touch and turn to greet you. They have noted that subtle actions help bring robots to life. Artificial eyes that move and blink, slight chest movements that simulate breathing, and man-made muscles to change facial expressions, are all attributes socially acceptable robots of the future must have.

Scientists have also been working on robotic pets. With no feeding required, a robotic dinosaur that recognizes sounds, sights, and touch, craves affection, and explores its environment is an alternative that any child would take to. Wouldn't you?





Let's Practice!

What will robots be like in the future?

Instructions: Circle all the phrases in the sentences that talk about what a robot may have, may do or may not do in the future.

- 1) Robotic engineers have been crafting robots to look, feel and act more human.
- 2) They have been working toward creating robots that sense your touch and turn to greet you.
- 3) These robots have artificial eyes.
- 4) They simulate breathing.



Let's Practice!

The robots look so life-like!

Instructions: Organize the mystery sentence.

1. who / a / . / was / appearance / in / humanoid / built / Westinghouse / robot /

2. The / photoelectric / routines / robot, / which / relays, / and / performed / . / 26 / cells
/ on / relied /

3. we / as / warm / machines / that / cold / makes / see / . / movement / Subtle / to /
easier / to / it /

4. the / sense / planned / for / and / are / turn / touch / Robots / you / future. / your /
that / greet / to /





Extension Activity

Write a letter for a contest.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Write a letter to win an expensive robotic pet for yourself or for someone you know.
- To win you have to say how long you have been interested in robots.
- You should tell them how you have always wanted one and why.
- Read your letter aloud to someone.

Self-Assessment

Yes

No

I write about robotic technology.

☐ ☐

I use recent actions and routines to talk about how long and why I want a robotic pet.

☐ ☐

I share my letter with someone.

☐ ☐


Target Words

- **potential** (adjective): showing possibility
e.g. Research into how robots function and their potential uses has grown.
- **substantially** (adverb): a great deal
e.g. Robotic research has grown substantially.
- **photoelectric** (adjective): using electricity produced by light
e.g. The robot relied on motors and photoelectric cells.
- **relay** (noun): a switch in an electrical circuit
e.g. The relays helped send the electricity.
- **to craft** (verb): to make objects with skill
e.g. Robotic engineers have been crafting robots to act more human.
- **to sense** (verb): to feel
e.g. They have been working toward creating robots that sense touch.
- **subtle** (adjective): not obvious
e.g. They have noted that subtle actions help bring robots to life.
- **facial** (adjective): of the face
e.g. In the future, robots will have facial expressions.





Learning Points

Present Perfect Continuous vs. Present Perfect

Both the Present Perfect and the Present Perfect Continuous are used to express that an action began in the past and is still going on or has just finished.

We use the Present Perfect

- to talk about something that started in the past and continues to the present
- to talk about something we have done several times in the past and continue to do
- when we talk about our experience up to the present
- to talk about something that happened in the past but is important at the time of speaking

We use the Present Perfect Continuous

- to describe something changing or developing up to now

Sometimes there's no difference in meaning between the two tenses. This is especially true with the verbs *live*, *work* and *study*.

Examples:

- *Research into how robots function has grown substantially since the early 20th century.*
- *Scientists have noted that subtle actions help bring robots to life.*
- *Robotic engineers have been crafting robots to look, feel and act more human.*
- *They have been hoping this will make it easier for us to warm to machines.*
- *They have been working toward creating robots that sense your touch.*





Learning Points

Subject and Object Relative Clauses

You put a relative clause immediately after the noun it talks about.

You use a relative clause with *which* for things to make the thing you are talking about clearer.

In spoken English and written English *that* often replaces *which* or *who*. *Which* is generally more formal than *that*.

Example	Clause	Relative Connector	Type of Relative Clause
<i>Westinghouse built a robot who was humanoid in appearance for the World's Fair.</i>	who was humanoid in appearance	who	subject
<i>The robot, which relied on photoelectric cells and relays, performed 26 routines.</i>	which relied on photoelectric cells and relays	which	subject
<i>Subtle movement makes it easier to warm to machines that we see as cold.</i>	that we see as cold	that	object

